

1 SAFETY PRECAUTIONS

1.1 Summary

This chapter describes the proper use of the warning terms, precautions, and warning labels that are attached to the device. The warning terms must be observed by the owners and personnel who install, operate, maintain, and inspect this device.

WARNING

- **When installing, operating, maintaining, and inspecting the device, always follow the instructions and warnings found within this manual.**
- **When there are any questions or when anything is unclear, contact the distributor or Ishida customer support staff. Do not proceed with any operations until the required instructions are given.**

1.2 Warning Indications-Types and Definitions

The warning instructions contained in this manual, as well the indications on the labels attached to the machine, are ranked into three categories according to the level of hazard involved. It is important that these indications be fully understood and complied with.

Table 1-1

Notation	Description
 DANGER	If this hazard is not avoided, death or serious injury will probably result. This indicates an imminent danger, and extreme caution must be exercised to prevent a mishap.
 WARNING	If this hazard is not avoided, there is a possibility of death or injury resulting.
 CAUTION	If this hazard is not avoided, there is a possibility that light or moderate injury may result. It may also indicate that a possibility of damage to device exists.
NOTE	Used for cautionary or important information to be noted or emphasized.
TIP	Used to indicate reference information for operations or information which helps understanding of the device.

1.3 General Precautions

To prevent hazards to person and damage to properties, items described in this chapter must be strictly observed.

In addition, notes which are specific to this equipment as described in section "1.4 Special Precautions" must be strictly observed.

< Safety precautions regarding equipment installation >

Table 1-2

Degree of hazards	Possible hazards and its prevention methods
 DANGER	Electrical wiring work must be performed by a legally qualified person. Electrical wiring work on the equipment must be performed by a legally qualified person. It is the violation of applicable laws and regulations for a person who is not legally qualified to perform electrical wiring work of the equipment. Faulty electrical wiring work may cause fire, electric shock, ground leakage, etc.
 DANGER	Ground the equipment. The equipment must be connected to protective ground. Securely connect the equipment to protective ground by having a legally qualified person perform the electrical wiring work. Improper grounding may result in an electric shock
 DANGER	Connect the equipment to a power supply with the voltage and frequency as indicated in the specifications and on the device plate. Connecting to a power supply with different voltage or frequency may result in equipment malfunction.
 DANGER	Avoid multiple-tap connection to a single outlet for power supply cable and other cables. Otherwise it may cause burning or start a fire.
 DANGER	Do not pour water on the equipment. Do not install the equipment at a location exposed to moisture or condensation. Except for water-resistant equipment, exposing to water will result in loss of performance, reduction in functions, equipment failure, or electric shock.
 DANGER	Install the equipment at a solid and level location. Installing the equipment at an unstable or tilting location may cause loss of performance, functions, or roll-over of the equipment, resulting in injury.

Table 1-2

Degree of hazards	Possible hazards and its prevention methods
 WARNING	Install the equipment with sufficient distance maintained from other electronic devices. Otherwise malfunction of the equipment or other electronic devices may occur.
 WARNING	Maintain necessary space for operation and maintenance. Otherwise an operational error or injury may occur.
 CAUTION	Do not use the same power supply for devices which may result in malfunction or damage. Otherwise the machine may malfunction or break down.

< Safety precautions regarding equipment operation >

Table 1-3

Degree of hazards	Possible hazards and its prevention methods
 DANGER	Do not touch the equipment with wet hand. Doing so may result in electric shock.
 DANGER	Do not step on or pull wet or damaged cables of the equipment. Doing so may result in electric shock, fire, or injury.
 DANGER	Do not put any object on the power supply cable and protect it from being damaged by the equipment or device placed on the cable. Damaging the cable could result in electric shock or fire.
 DANGER	Do not try to disassemble, repair, or modify the equipment. Do not try to disassemble, repair, or modify the equipment with tools such as screwdriver or spanner unless indicated in the instruction manual. Doing so may result in electric shock, fire or injury. All repairing the device can be performed by Ishida service department. Contact the distributor or Ishida Customer Support.
 DANGER	Be sure to check that there is no damaged parts or missing parts before starting operation. Otherwise parts fragment may become mixed into a product handled by the equipment, resulting in a foreign matter contamination accident. Electric shock, fire or injury may result.

Table 1-3

Degree of hazards	Possible hazards and its prevention methods
 DANGER	Start operation after checking that nothing is placed on the equipment. The object placed on the equipment may become mixed together with product handled by the equipment, resulting in a foreign matter contamination accident. Electric shock, fire or injury may result.
 DANGER	Operate the equipment with all the covers and doors closed. Otherwise may result in loss of performance, reduction in functions, or injury.
 DANGER	Do not insert your hand or finger in the gap of the cover or door of the equipment. Doing so may result in loss of performance, reduction in functions, or injury.
 DANGER	Conduct maintenance and cleaning of the equipment. Make sure to conduct maintenance and cleaning of the equipment according to the instruction manual. Not doing so may result in loss of required hygienic level of foods handled by the equipment.
 WARNING	The equipment must be operated by a person who has read and understood the instructions in the manual. Make sure to fully understand precautions, methods on operation and maintenance according to the instruction manual before operating the equipment. Otherwise electric shock, injury, fire, damage to the equipment or food poisoning caused by foods handled by the equipment, may occur.
 WARNING	Operate the equipment with appropriate clothes. Be sure to fasten or zip up any fasteners on the clothes. Put on non-slip shoes and tightly tie the shoe laces. Remove any accessories that you are wearing. Remove tie and strap, or fasten them tightly with a clip. Cover your hair entirely with a hat or hair cap. Disregarding the above instructions may result in hair or fibers mixed into the product handled by the equipment and cause a foreign matter contamination accident. In addition, it may result in an injury caused by entanglement of foreign matter in the equipment.
 WARNING	Make sure that there is no other person around the equipment before operating the equipment. Injury may occur when trying to operate the equipment suddenly while other person is near the equipment.
 WARNING	Do not touch the operating parts of the equipment unless indicated in the instruction manual. Doing so may result in loss of performance, reduction in functions, or injury.

Table 1-3

Degree of hazards	Possible hazards and its prevention methods
 WARNING	Use fingers to operate the control panel. Operating the control panel with other than fingers may result in damage of the control panel.
 WARNING	Do not ride on, lean against, or sit on the equipment. Doing so may result in loss of performance, reduction in functions, or injury.
 WARNING	Use a stool or step ladder in accordance with its instruction manual and as indicated. When using a stool or step ladder, make sure to follow the instructions in the manual as indicated. Otherwise it may result in injury.
 WARNING	Be sure to wait at least 5 minutes after turning off the power supply before conducting maintenance or inspection of the equipment. Otherwise may result in electric shock.
 WARNING	Do not conduct any maintenance or inspection that is not described in the instruction manual. Doing so may result in electric shock, fire or injury.
 WARNING	Do not turn off the main power supply switch during operation except in emergency. Turning off the main power switch of the equipment during operation may delete the configuration and summary data.
 CAUTION	Do not allow foreign objects or liquid such as insecticide to enter the protected sections of the equipment such as the main body, terminal box, and motor box. They may lead to equipment malfunctions or breakdown.

1.4 Special Precautions

This section describes special safety and sanitary precautions for this device which, in addition to the previously mentioned precautions, should be carefully observed.

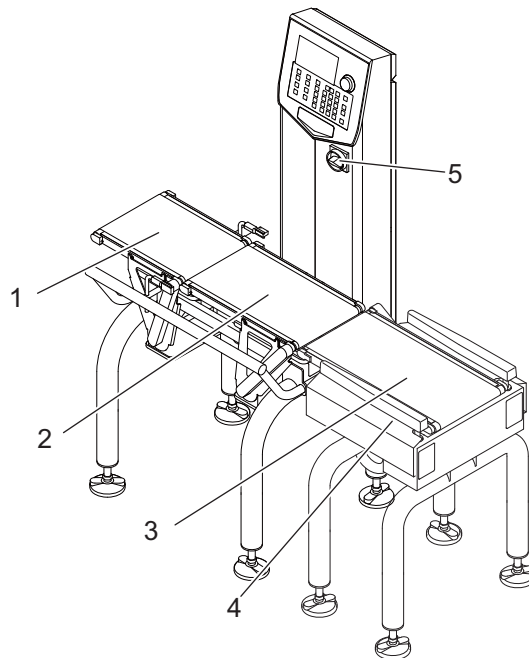


Fig. 1-1

Table 1-4

Degree of hazards	Possible hazards and its prevention methods	Fig. No.	Chapter
 WARNING	Do not place your hand between conveyors. Otherwise, the conveyor may entangle your hands, which may cause an injury.	1, 2, 3	"2.2"
 WARNING	When cleaning the device, the operator must turn OFF and lock the main power switch and keep the key in his possession during the work to prevent other personnel from starting the device accidentally, which may cause an electrical shock or injury.	5	"1.6"
 CAUTION	Be sure that a chute of the rejecter device is installed before operating the device.	4	—
 CAUTION	(When arm rejecter is used) When supplying air, do not place your hand or an object on the conveyor. The arm may suddenly move and you may get injured.	1, 2, 3	—
 CAUTION	During operation or just after stopping the device, do not touch the moving parts. Oil seal to protect the bearing is very hot due to frictional heat. Such temperature rise of oil seal is not abnormal.	1, 2, 3	—

Table 1-4

Degree of hazards	Possible hazards and its prevention methods	Fig. No.	Chapter
 CAUTION	Route the wiring in the way that power voltage fluctuation due to load change does not exceed $\pm 10\%$. Failure to do so may cause malfunction or damage to the device.	—	"3.2"
 CAUTION	When relocating the device, consult the distributor or Ishida customer support. Inappropriate installation may result in inaccurate weighing.	—	"3"
 CAUTION	Do not apply an excessive load to weigh conveyor. Doing so may damage the weighing part.	2	—
 CAUTION	When the cleaning procedures specified, wipe with a dry cloth. Failure to do so may cause malfunction or damage to the device. For unpacked food, the risk of an unhygienic condition may occur through the usage of the device. To prevent such risk, clean the device properly depending on product type and processing methods. For cleaning methods, read and thoroughly understand Chapter 8 and follow the instructions.	—	"7"
 CAUTION	Install this device in a location fulfilling the following conditions. Failure to observe the instructions may cause breakdown or malfunctions of the device. • Do not place in a location subject to dust, vibration, direct sunlight or extreme heat. • Do not place the device near equipment which generate electromagnetic interference. • Ambient temperature range: 0°C to 40°C • Ambient Humidity: 35% to 85% (no condensation) • Do not water the upper portion than the legs. • A hard level surface with minimum vibration. When a metal detector is connected, install the device so that the following conditions are also fulfilled. • Ensure that there is no moving metal objects near the device other than Ishida devices. • Do not place the device near metal detectors operating at identical frequency. • Ensure that the metal detector is fixed securely on the device.	—	"3"

Table 1-4

Degree of hazards	Possible hazards and its prevention methods	Fig. No.	Chapter
	The conveyor is mounted by the spring force. When mounting or removing the conveyor, take extreme caution to the following: • Removing the conveyor: If the shaft is released from the hook, the moving part at the conveyor end is loosened and your hand may be caught in the conveyor frame. Take caution not to hold near the moving part. • Mounting the conveyor: When pushing the shaft at the conveyor end in the hook, hold the conveyor bracket with a finger taking care not to be pinched and push in carefully.	1, 2, 3	"7.3"
	(When a metal detector is connected) When the setting value was changed during the production, stop the production to confirm the detecting sensitivity. Otherwise, the metal detector will not operate accurately for a certain period after the change of setting value during the production.	—	—
	(When a metal detector is connected) <Information about detecting sensitivity> • Type of metal and detecting sensitivity The detective size of metal depends on the type of metal, such as steel, copper, aluminum, and stainless, because the metal detector's response to the metal and detection principle depend on the type of metal. Be sure of the following when confirming the detecting sensitivity. • Shape of metal and detecting sensitivity The detecting sensitivity is indicated by the diameter of the smallest metal ball to be detected. The sensitivity of metal bar depends on the angle when it passes through the metal detector. Note that non-magnetic metal is not detected due if the diameter of cross section is less than that of test piece due to the principle, regardless its length. • Product effect and detecting sensitivity Depending on the character of the product, the metal detector responses the product as if it is a metal. This effect is called "Product effect". Product effect depends on each products and the product effect for the same product depends so much on its state such as temperature, water or oil content. When the product effect is large, the detecting sensitivity needs to be decreased to prevent the masticating due to the product effect. In such case, note that the sensitivity to metal is also decreased. • Transition of product effect and detecting sensitivity Product effect changes according to the change of the state over time. To keep detecting sensitivity high, control the state of product so that the product effect is kept as small as possible.	—	—

1.5 Warning Labels

Warning labels which indicate points requiring particular caution are attached to the weigher at certain locations. Please take sufficient time to familiarize yourself thoroughly with the meanings and positions of these labels.

1.5.1 Warning Label Handling

- Check if all the warning labels are legible. If any letters or illustrations are unclear, clean or replace such labels.
- Clean the warning labels using a cloth, water, and neutral detergent.
Do not use organic solvents or gasoline.
- If the warning labels are damaged, lost, or illegible, replace the labels.
Check the machine type (model) and machine identification, and contact the distributor or Ishida customer support.

1.5.2 Warning Label Location

The diagram below shows the location of the warning label.

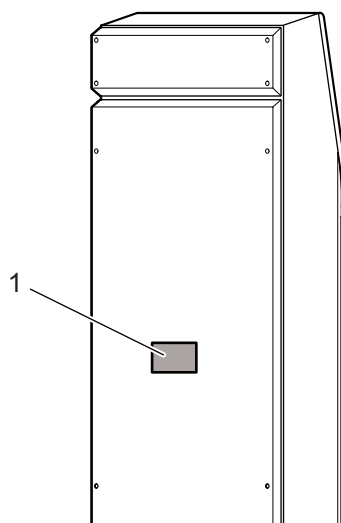


Fig. 1-2

Table 1-5

No.	Location	Label	Meaning of hazard	Prevention Method of hazard
1	Rear of the machine		ELECTRIC SHOCK	Disconnect from main power before opening.

1.6 Drive Power Shutdown and Indication

CAUTION

- When performing maintenance or inspections, turn OFF the power supply to secure the safety of operator.
- Note that if the main power switch is turned OFF, the power is supplied to the main body circuit to the main power switch.

When performing maintenance or inspections, the following procedure is recommended to prevent the other operators from accidentally turning ON the power supply.

- Lock the main power switch in the OFF position.
- Create an accident prevention tag and place it at power shut-off location.

The figure below shows the placement of a main power switch lock and accident prevention tag.

NOTE

- A main power switch lock and accident prevention tag are to be prepared by customers.

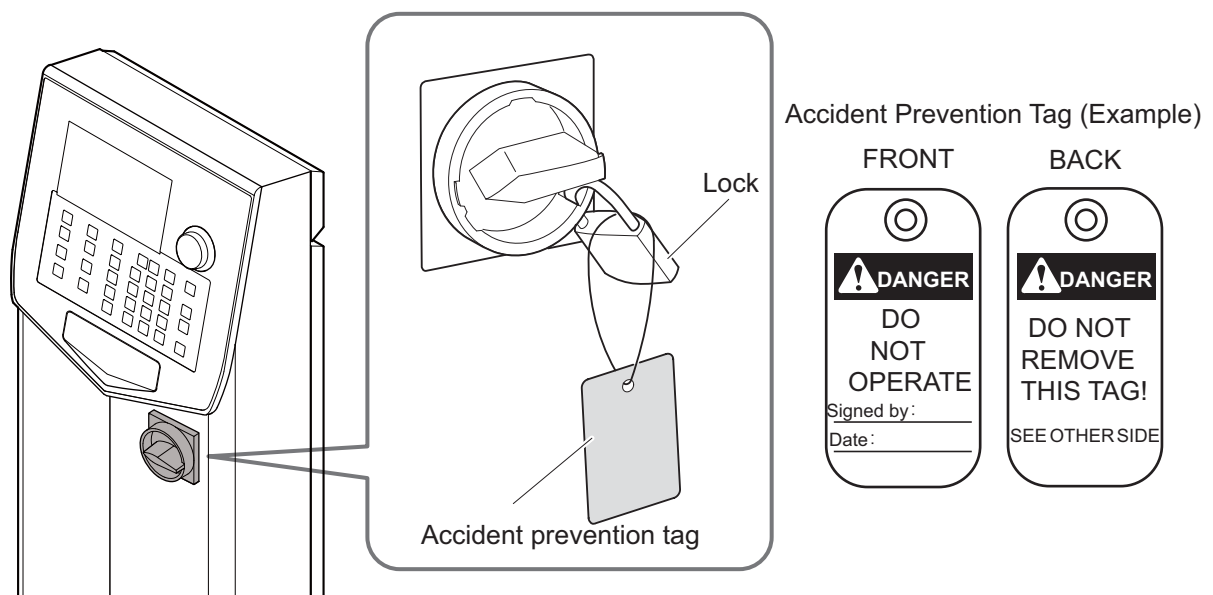


Fig. 1-3